TITAN molecular prediction report

24 August 2026

Contents

Research-use molecular profile

COAD - TCGA-AA-3972

Models applied	Continuous	Binary	Slides pooled
20	10	10	1

This report applies models trained in TCGA and **not externally validated**. It is a discovery/research output, not a diagnostic report. Binary LDA scores are uncalibrated. Their displayed TCGA out-of-fold score ranks describe relative position only and must not be read as probabilities.

Case context (not used as model input)

Shared selection features. Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins.

Pathology summary. The sigmoid colectomy contained an ulcerated grade-2 colorectal adenocarcinoma extending through the bowel wall into adjacent mesocolic fat (pT3). The proximal and distal resection margins were uninvolved; the supplied slide summary did not state a nodal category.

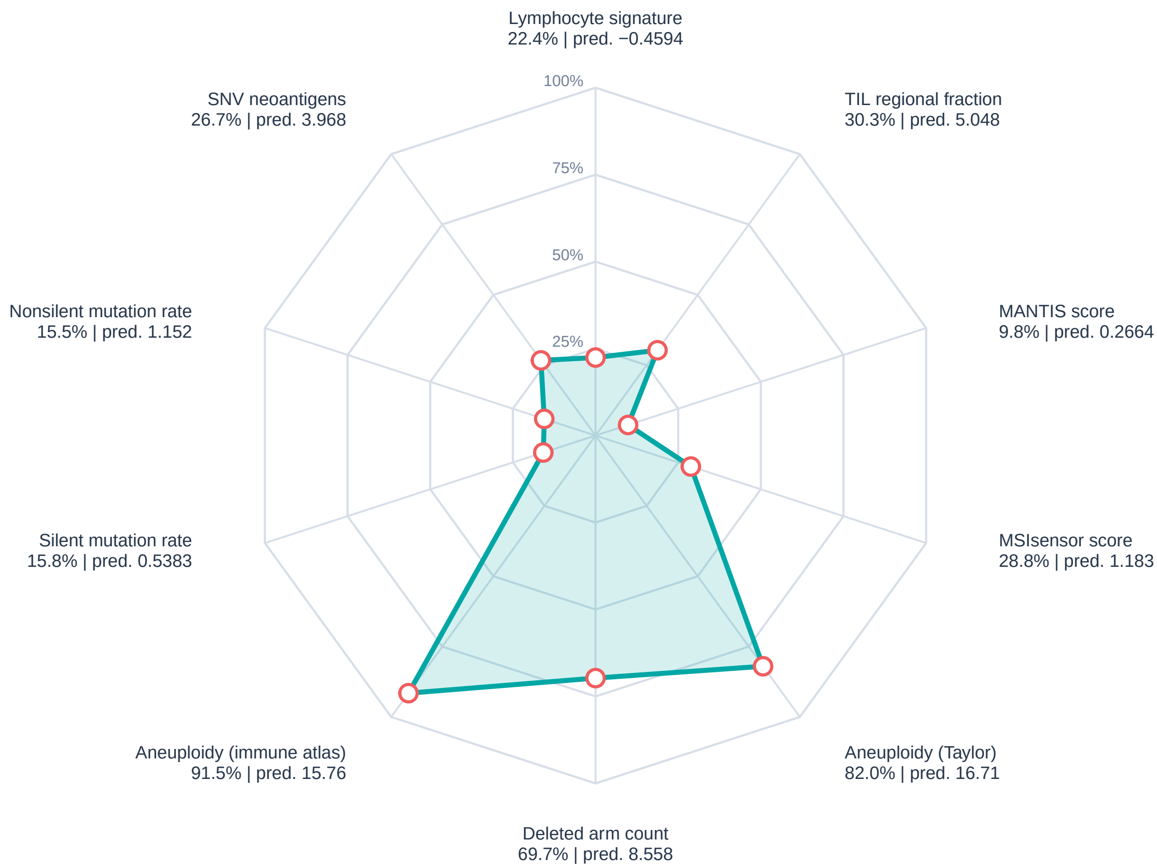
Treatment and follow-up. NCI GDC records list capecitabine and oxaliplatin beginning on day 60, with the recorded outcome 'Progressive Disease'; recurrence was recorded on day 1216. Later records list fluorouracil, leucovorin, oxaliplatin and bevacizumab through day 1369, also with progressive disease. Follow-up on day 1551 was 'with tumor'; the recorded vital status was alive.

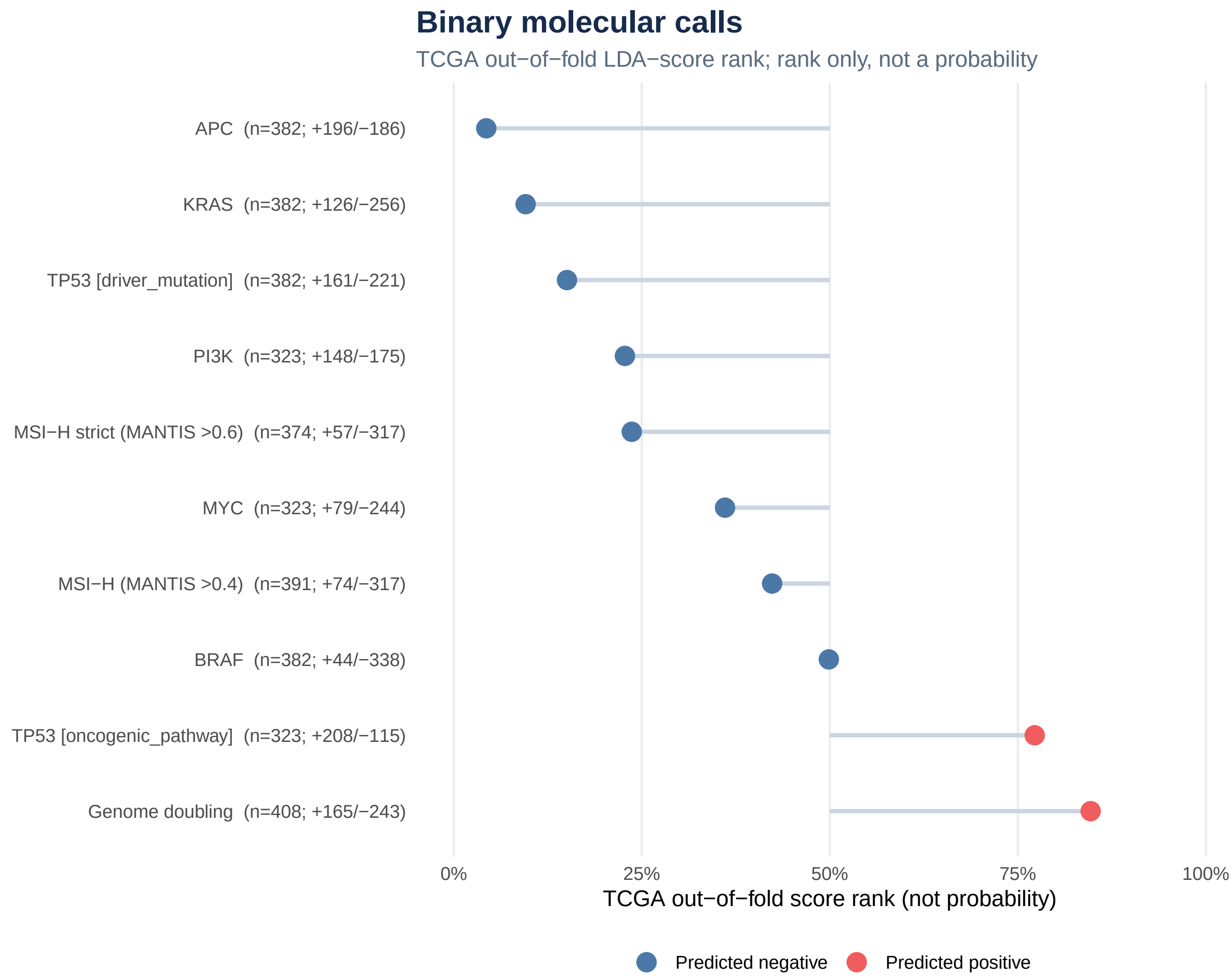
Clinical-context source. Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3). Treatment and follow-up fields were retrieved from the public NCI GDC Cases API (https://api.gdc.cancer.gov/cases) on 16 August 2026.

The pathology and treatment fields are descriptive public metadata; they were not predictors, training targets, or validation outcomes for TITANPred. No treatment-effect inference should be drawn from these examples.

Continuous molecular prediction radar

Corner labels: TCGA reference percentile | original model prediction





Continuous predictions

Family	Endpoint	Prediction	Reference percentile	Category
thorsson	Aneuploidy Score	15.76	91.5%	Moderate effect
microsatellite_instability	MANTIS score	0.2664	9.8%	Moderate effect
thorsson	Nonsilent Mutation Rate	1.152	15.5%	Moderate effect
thorsson	Silent Mutation Rate	0.5383	15.8%	Moderate effect
aneuploidy	Aneuploidy score	16.71	82.0%	Moderate effect
thorsson	Lymphocyte Infiltration Signature Score	-0.4594	22.4%	Moderate effect
thorsson	SNV Neoantigens	3.968	26.7%	Moderate effect
microsatellite_instability	MSIsensor score	1.183	28.8%	Moderate effect
aneuploidy	Deleted arm count	8.558	69.7%	Moderate effect
thorsson	TIL Regional Fraction	5.048	30.3%	Higher effect

Binary predictions

Endpoint	Predicted class	LDA score	TCGA OOF score rank (not probability)	Training n	Positiv
Genome doubling	1	1.557	84.7%	408	16
APC	0	-3.918	4.3%	382	19
BRAF	0	-5.412	49.9%	382	4
KRAS	0	-4.625	9.6%	382	12
TP53 [driver_mutation]	0	-3.37	15.1%	382	16
MSI-H (MANTIS >0.4)	0	-4.847	42.3%	391	7
MSI-H strict (MANTIS >0.6)	0	-9.961	23.7%	374	5
MYC	0	-2.331	36.1%	323	7
PI3K	0	-1.268	22.8%	323	14
TP53 [oncogenic_pathway]	1	1.99	77.2%	323	20

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.4%.
- TITAN input schema: 768 named features (`titan_000`–`titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- Intended use: research only; TCGA discovery atlas without independent external validation.

Endpoint interpretation and target sources

For continuous endpoints, the percentile locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, the displayed value is the rank of the uncalibrated LDA score among repeated TCGA out-of-fold scores. It is not a probability, confidence, or clinical-risk estimate. Training denominators and repeated-CV performance are shown beside each binary call; rows marked for limited class size require particular caution.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **BRAF.** Binary prediction of a qualifying protein-altering PASS mutation in BRAF; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **TP53 [driver_mutation].** Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **MSI-H (MANTIS >0.4).** Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.
- **MSI-H strict (MANTIS >0.6).** Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.

- **MYC.** Binary prediction that the MYC oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **PI3K.** Binary prediction that the PI3K oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **TP53 [oncogenic_pathway].** Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score.** Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **Deleted arm count.** Number of chromosome arms classified as deleted; larger values indicate a greater arm-level deletion burden. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score.** Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale:* source endpoint units *Model-target source:* Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score.** Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale:* source endpoint units *Model-target source:* Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **Aneuploidy Score.** Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Lymphocyte Infiltration Signature Score.** Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Nonsilent Mutation Rate.** Rate of coding mutations that alter the encoded protein. TITANPred reports the model output on the analysis log(1+x) scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **SNV Neoantigens.** Predicted neoantigen burden arising from single-nucleotide variants. TITANPred reports the model output on the analysis log(1+x) scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Silent Mutation Rate.** Rate of coding mutations that do not change the encoded amino acid. TITANPred reports the model output on the analysis log(1+x) scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction.** Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.